DP Biology – A2.1 (HL only)

Origins of cells

- How could life have begun from non-living matter?
- What stages might lie between simple chemicals and the first cells?

HL statements A2.1.1–9. Textbook pp. 34–47.

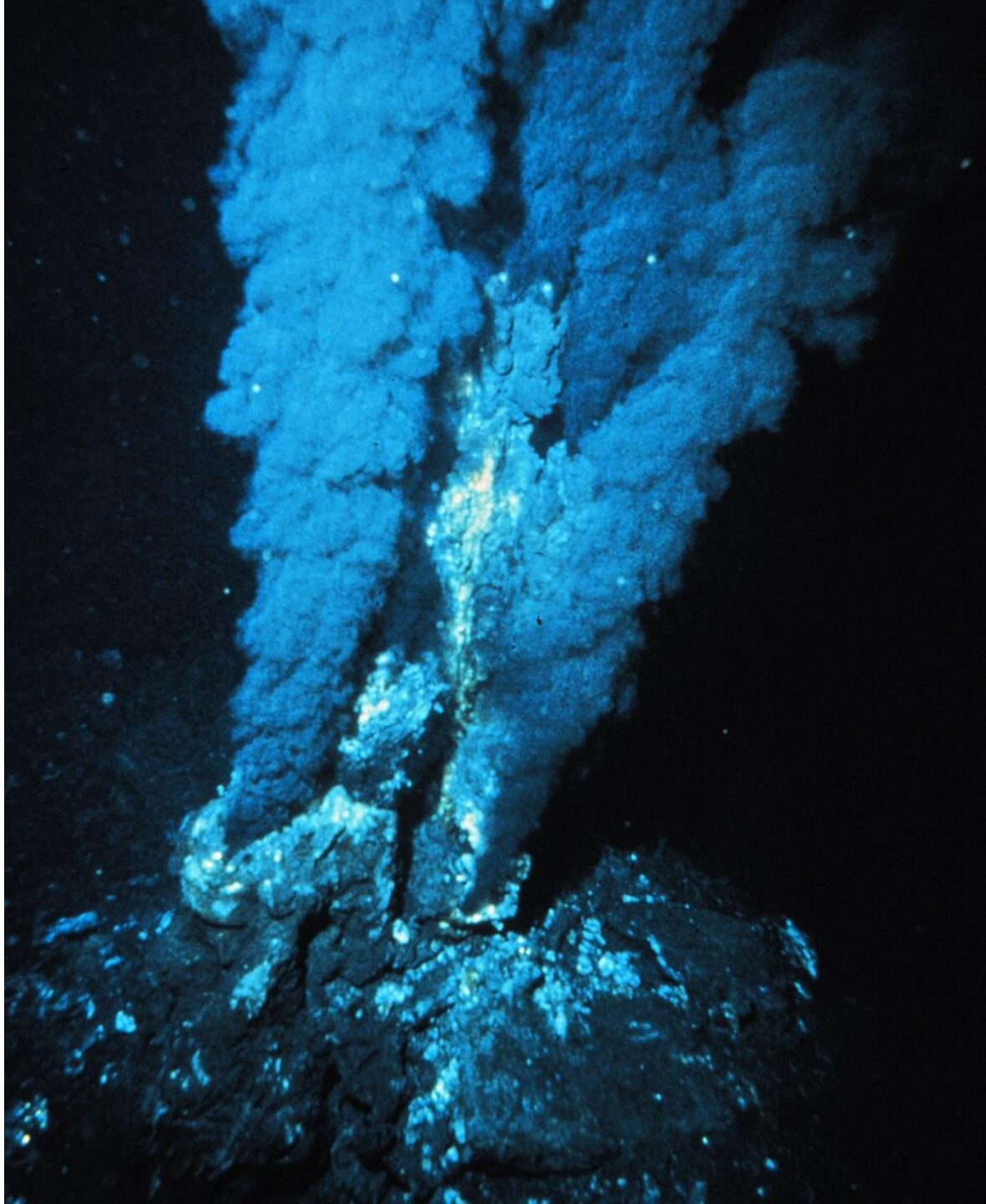


Image: a black smoker on the Mid-Atlantic Ridge. P. Rona, NOAA, public domain.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	A2.1.1–3	Describe the early Earth, explain why cells are the smallest self-sustaining units, and why their origin is hard to explain
2	A2.1.4–6	Evaluate the Miller–Urey experiment, and explain spontaneous vesicles and the RNA world
3	A2.1.7–9	Outline the evidence for LUCA, how early life is dated, and the case for hydrothermal vents

The big picture: from chemistry to cells



No one saw life begin, so scientists test each step with **experiments**, **fossils** and **genes**.

Big question

If cells only come from cells, how did the first cell arise?

Conditions on the early Earth

A2.1.1



Lava fountain, Hawaii. J. D. Griggs, USGS, public domain.

Feature	Early Earth	Today
Free oxygen	Almost none	About 21%
Ozone layer	Absent	Present
CO_2 and methane	Much higher	Traces
Temperature	Much higher	About 15 °C
UV at surface	Very high	Mostly absorbed

Cause and effect

No oxygen meant no ozone, so UV reached the surface. Volcanoes, lightning and UV supplied energy.

Carbon compounds before life

A2.1.1

Energy could join simple gases into **amino acids**, **sugars** and **bases**, with no cells or enzymes.

This does not happen today: oxygen and living things destroy the products.



Kilauea eruption. Hawaii Volcanoes NP, public domain.

Discuss

UV damages large molecules, yet may help RNA form. Was UV more help or harm?

Cells: the smallest self-sustaining units

A2.1.2

Membrane	Genes	Metabolism
Keeps its chemistry separate	Pass on information	Release energy and grow

A cell carries out **all** life processes on its own. Its parts, such as a ribosome, are not alive alone.

Activity

Test a cell, a ribosome, a virus and a growing salt crystal against the life processes. Which are alive?

Why viruses are not considered living

A2.1.2

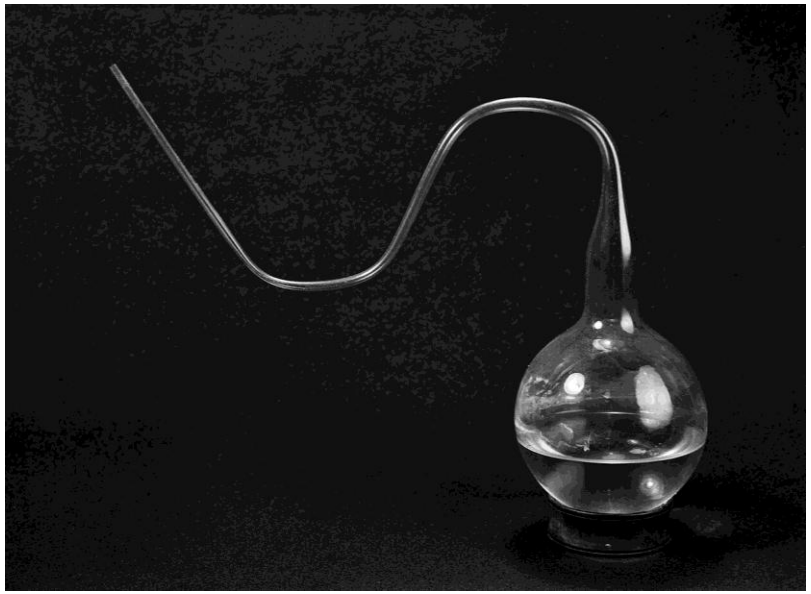
Feature	Cell	Virus
Made of cells	Yes	No
Own metabolism and ATP	Yes	No
Own ribosomes	Yes	No
Reproduces	By dividing	Only inside a host
Evolves	Yes	Yes

Key idea

Viruses have genes and evolve but cannot sustain themselves (A2.3).

Cells only come from cells

A2.1.3



Pasteur's swan-necked flask. Wellcome Collection, CC BY 4.0.

Pasteur's boiled broth in a **swan-necked flask** stayed sterile: air entered, but microbes settled in the bend.

So cells come only from existing cells.

The challenge

Yet the very first cells must have formed from non-living matter. Cells are so complex that this is one of biology's hardest problems.

Four requirements for the first cells

A2.1.3

Catalysis

Reactions fast enough for metabolism

Self-replication

Molecules that copy themselves

Self-assembly

Structures such as membranes forming alone

Compartments

A boundary separating inside from outside

Hypothesis	Idea
Protocell first	Membrane compartment first; genes later
Gene first	A self-replicating RNA first
Metabolism first	Reactions on mineral surfaces first

None is proven: conditions cannot be recreated and protocells left no fossils.

Practice: early Earth and life

A2.1.1–3

Explain

Why strong UV reached the early surface.

Classify

A ribosome: alive or not? Why?

Explain

Why the first cell is hard to explain.

Use

ozone, self-sustaining, spontaneous generation, compartment

Check-in: class 1

A2.1.1–3

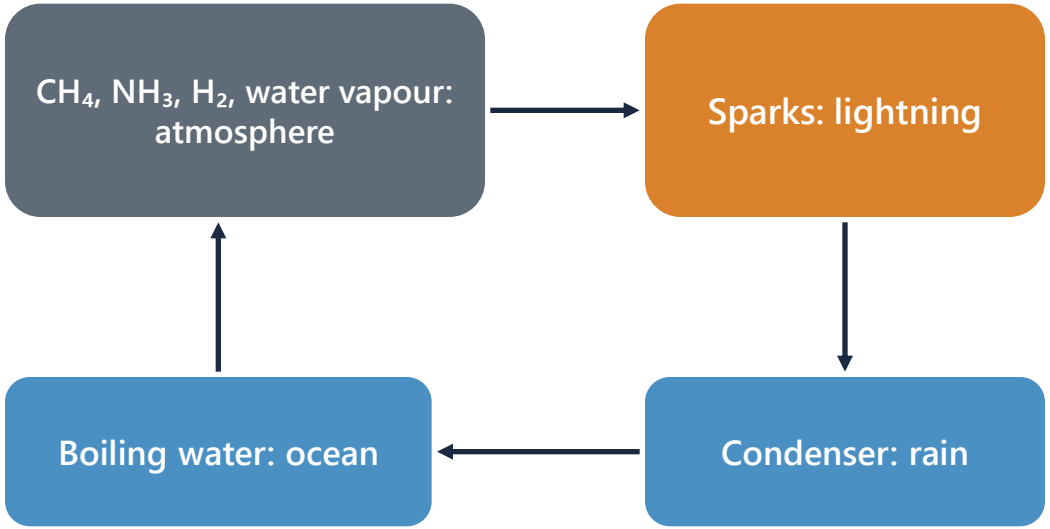
1. Describe two ways the early Earth differed from today.
2. Explain why a cell is self-sustaining but a ribosome is not.
3. Give two reasons viruses are not considered living.
4. Outline Pasteur’s evidence against spontaneous generation.
5. List the four requirements for the first cells.

How to use this

Answer without your notes, then check against slides 3–10.

The Miller–Urey experiment, 1953

A closed glass loop: gases pass the sparks, then condense and collect products.



Results

After a week the water was brown and held several amino acids. Later repeats made sugars, fatty acids and bases: building blocks form without life.

Evaluating Miller and Urey

A2.1.4

Strengths	Limitations
Closed, controlled, repeatable	Early atmosphere probably had less methane, more CO_2
Organic molecules made without life	Sparks used; UV probably mattered more
Later versions agree	Monomers are far from polymers or cells

Activity

Write an evaluation: what it showed, two limitations and their effect, one improvement.

Working with data: what forms, and how fast?

A2.1.4

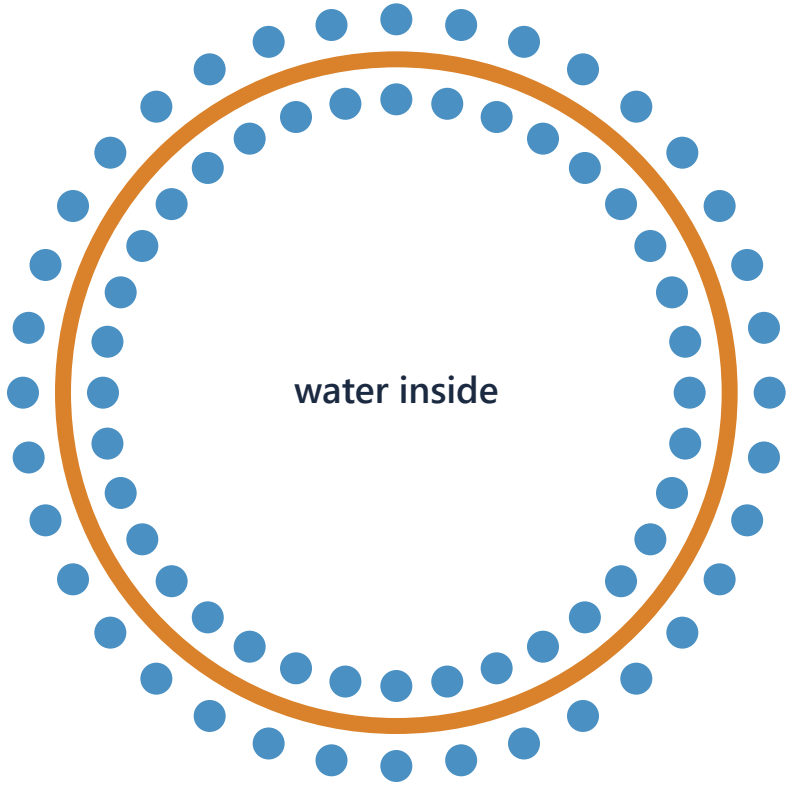
Amino acids detected in a spark experiment after different run times. Invented values; realistic pattern.

Run time / days	Glycine / mg	Alanine / mg
1	2	1
3	9	5
7	18	10
14	20	11

1. Describe the trend.
2. Suggest why the yield levels off.
3. Suggest a control.
4. Why do glycine and alanine form most?

Vesicles that form by themselves

A2.1.5



blue heads face water; orange tails hide inside the bilayer

Fatty acids are **amphipathic**. In water they form bilayers that close into **vesicles**, **spontaneously**.

Fatty acids are simpler than phospholipids, so probably formed the first membranes.

Why it matters

A compartment keeps molecules together and lets inside chemistry differ from outside.

RNA as the first genetic material

A2.1.6

The problem

DNA needs enzymes to be copied, but enzymes are coded in DNA. RNA could do both jobs.

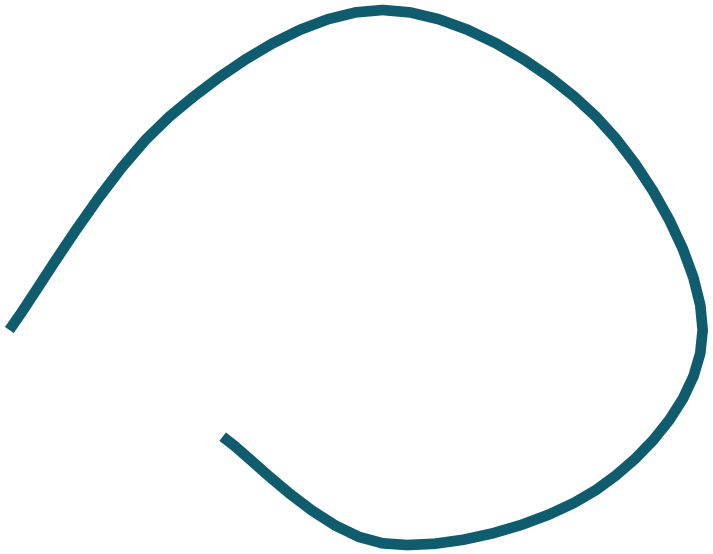
RNA can…	Evidence
Store information	Base sequence
Be replicated	Base pairing
Catalyse	Ribozymes

In ribosomes, **rRNA** catalyses peptide bonds: a relic of the RNA world.

DNA later took over storage because it is more stable.

One molecule, two jobs

A2.1.6



folded RNA: a ribozyme

Information

base sequence copied by pairing

Catalysis

folds into an active site

Think

Why is a molecule that can copy itself and catalyse reactions a good candidate for the first gene?

Practice: building blocks

A2.1.4–6

Evaluate

One limitation of Miller-Urey.

Explain

Why vesicles form without enzymes.

Explain

Why RNA may have come first.

Use
amphipathic, vesicle, ribozyme, replicate

Check-in: class 2

A2.1.4–6

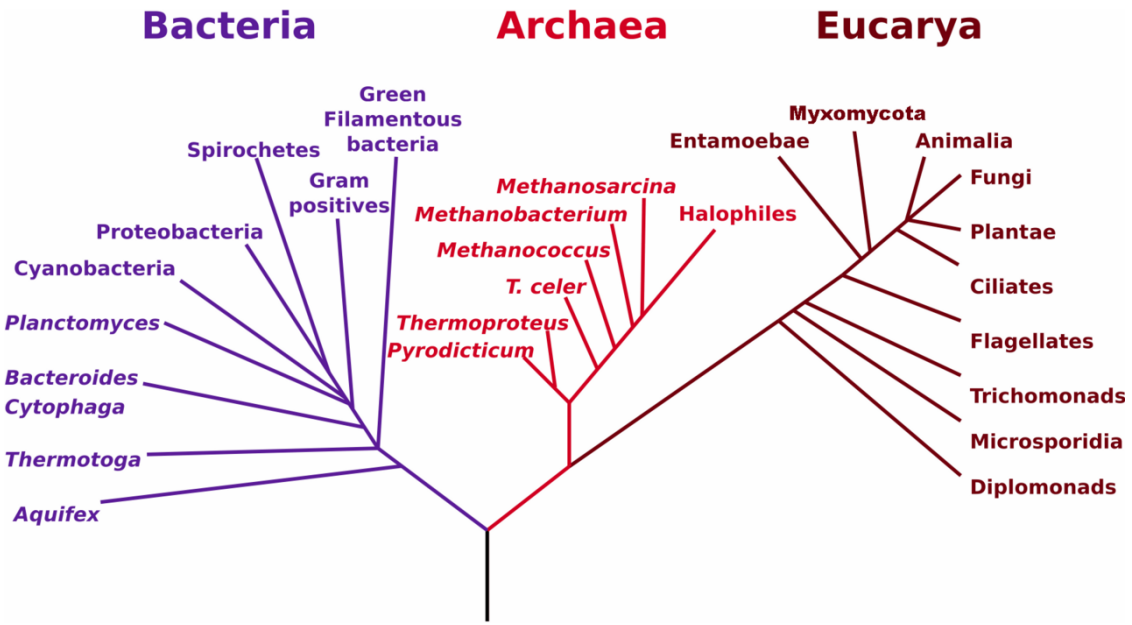
1. Outline what Miller and Urey showed.
2. Give two limitations of their experiment.
3. Explain how vesicles form.
4. State three things RNA can do.
5. Explain why rRNA in ribosomes supports the RNA world.

How to use this

Answer without your notes, then check against slides 12–18.

The last universal common ancestor

A2.1.7



Tree of life in three domains. MPF, public domain.

All life shares the **genetic code**, 20 amino acids and similar ribosomes.

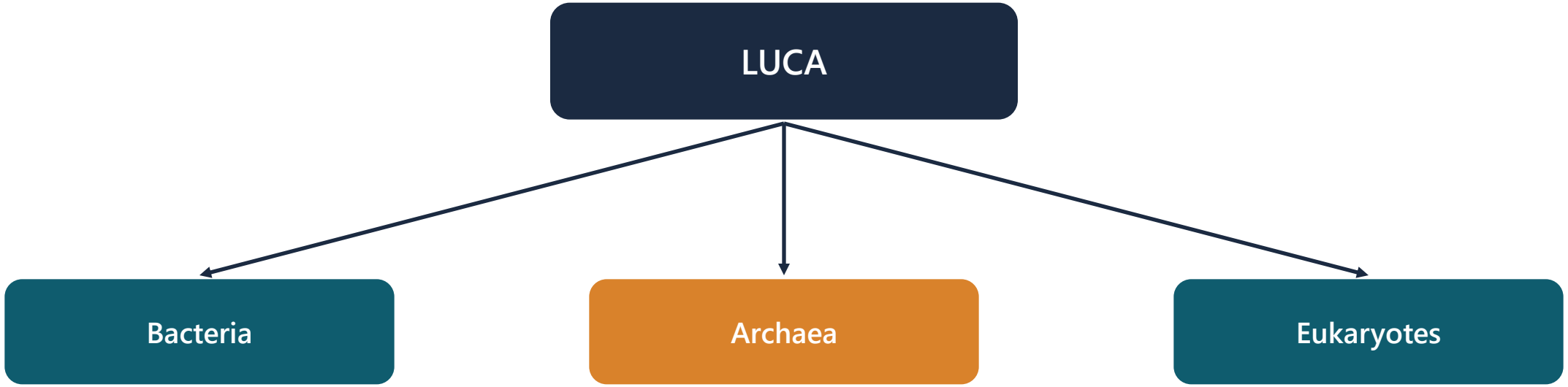
So all organisms descend from **LUCA**, about 4 billion years ago.

Other life?

Other early life may have died out competing with LUCA's descendants. LUCA was a cell, not a protocell.

Evidence for one ancestor

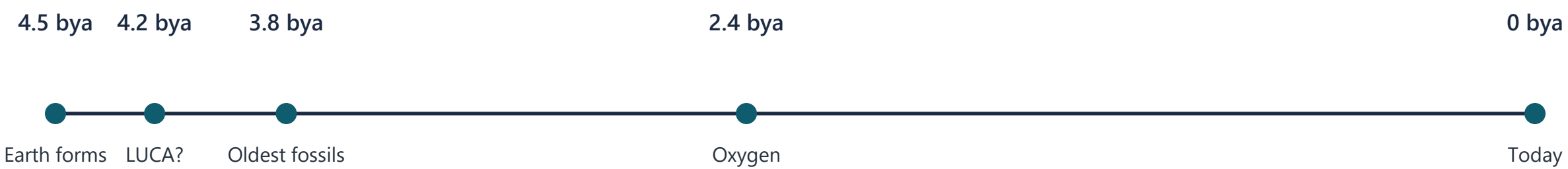
A2.1.7



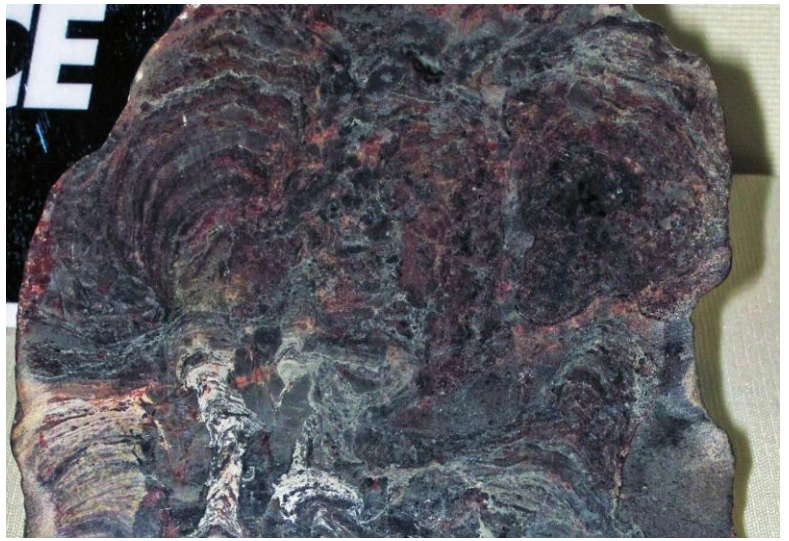
Shared by all	Why it points to one origin
Genetic code	Many codes possible; only one used
Ribosomes, ATP, DNA	Same machinery everywhere

Dating the first cells and LUCA

A2.1.8



Method	How
Fossils	Microfossils, stromatolites
Radiometric	Isotope decay dates rock
Molecular clock	Gene differences / mutation rate



Fossil stromatolite. James St. John, CC BY 2.0.

Working with data: dating the earliest life

A2.1.8

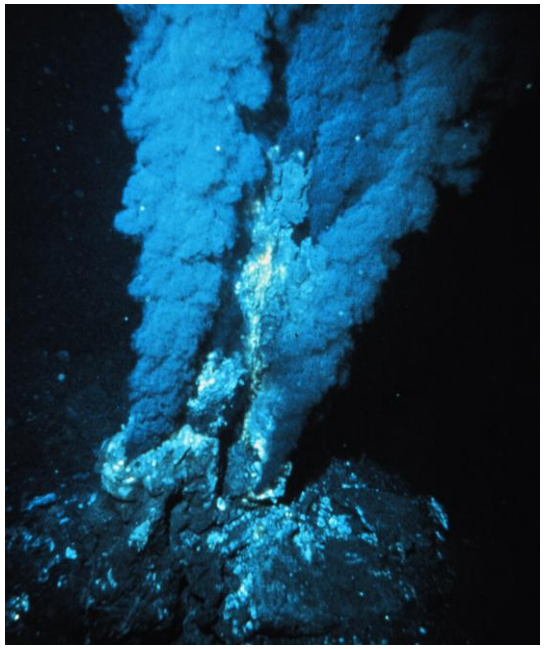
Approximate ages (rounded published estimates).

Event	Billion years ago
Earth forms	4.54
Oldest liquid water (zircons)	4.4
LUCA (molecular clock)	about 4.2
Oldest likely microfossils	3.7
Oldest accepted stromatolites	3.4

1. How long after Earth formed did LUCA live?
2. Why is LUCA older than the oldest fossils?
3. Why are these dates uncertain?

Did LUCA live near hydrothermal vents?

A2.1.9



Black smoker. P. Rona, NOAA, public domain.



Nuvvuagittuq rock, Quebec. Daderot, CC0.

Evidence	What it shows
Fossils	Vent-rock tubes at least 3.77 bya
Genes	LUCA suited heat, no O_2, hydrogen
Reverse gyrase	Enzyme of organisms above 80 °C

Vents as a cradle

Hot water rich in hydrogen and iron sulfide gives energy for joining carbon compounds. LUCA was probably anaerobic and heat-loving.

Practice: LUCA

A2.1.7–9

Explain

Why a universal code implies one ancestor.

Outline

One dating method.

Evaluate

The vent hypothesis.

Big question

Answer it now: how could the first cell have arisen, step by step?

Check-in: class 3

A2.1.7–9

1. Define LUCA.
2. Give two features shared by all life.
3. Outline how a molecular clock dates a common ancestor.
4. Give two pieces of evidence that LUCA lived near vents.
5. Explain why early life is hard to date.

How to use this

Answer without your notes, then check against slides 20–25.

Common mistakes to avoid

Not this	But this
LUCA was the first cell.	It was the last shared ancestor; others existed.
Miller and Urey made cells.	They made simple carbon compounds.
Viruses are the simplest cells.	Viruses are not cells.
Spontaneous generation happens today.	Cells now come only from cells.

Summary: origins of cells

A2.1.1–9

	Key idea
A2.1.1	Early Earth: no O_2, strong UV, energy sources
A2.1.2	Cells: smallest self-sustaining units
A2.1.3	Four requirements; no proven hypothesis
A2.1.4	Miller–Urey: building blocks form
A2.1.5	Fatty-acid vesicles self-assemble
A2.1.6	RNA world: genes and catalysts
A2.1.7–9	LUCA, dating, hydrothermal vents

Check that you can...

1. **Describe** the early Earth
2. **Explain** why cells are self-sustaining
3. **Evaluate** the Miller–Urey experiment
4. **Explain** vesicles and the RNA world
5. **Outline** evidence for LUCA and dating methods
6. **Evaluate** the vent hypothesis

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State two differences between the early atmosphere and today's.
2. Explain why RNA could have acted as both genes and enzymes in the first cells.
3. Give one piece of evidence that all life shares a common ancestor.

Image credits

Blacksmoker in Atlantic Ocean.jpg – P. Rona / OAR/National Undersea Research Program (NURP); NOA – Public domain

Pahoeoe fountain original.jpg – Jim D. Griggs, HVO (USGS) staff photographer[1][2] – Public domain

Kīlauea volcano eruption 20201220.jpg – Hawaii Volcanoes National Park – Public domain

Swan-necked flask used by Pasteur. Wellcome M0012521.jpg – Author not stated – CC BY 4.0

PhylogeneticTree.png – MPF – Public domain

Stromatolites (Bitter Springs Formation, Neoproterozoic, 850 Ma; Amadeus Basin, Australia) 4 (17164764207).jpg – James St. John – CC BY 2.0

Porpoise Cove (Nuvvuagittuq) banded iron formation, northern Quebec, metamorphosed 3.8 billion years ago, magnetite and quartz – Redpath Museum – McGill University – Montreal, Canada – DSC07889.jpg – Daderot – CC0

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